Question 1 (10 marks)
A company wants to move from transactional relationships to strategic alliances with suppliers.

a) Describe FOUR stages in the evolution of buyer–supplier relationships. (8 marks)

1. **Transactional stage**
   At this point, buyers and suppliers are mostly concerned with getting the best price and making quick purchases. They don't really talk to each other except to place orders, and the whole thing feels like a basic business deal. There's not a lot of trust or commitment, and it's easy for the buyer to switch to a different supplier if they want to, because they haven't invested much in the relationship. The focus is on short-term gains, rather than building a long-term partnership. As a result, the relationship remains fairly superficial, with little effort put into communication or collaboration beyond the basics of buying and selling.

2. **Transactional–operational stage (or cooperative stage)**
   The relationship becomes more structured, with regular orders and some basic performance monitoring. The buyer may start using simple KPIs such as delivery time and quality. There is still no deep partnership, but both parties begin to rely on each other more and may share basic information.

3. **Collaborative stage**
   As the relationship between the buyer and key suppliers evolves, they begin to collaborate more intimately, exchanging vital information on demand forecasts, production capacity, and potential quality concerns. This closer partnership often involves joint meetings to drive improvement, collective problem-solving, and the development of coordinated plans. The emphasis gradually shifts away from solely focusing on the price, instead embracing a more holistic approach that considers total cost and overall performance. By doing so, both parties can work together to optimize their operations, leading to enhanced efficiency and mutual benefits.

4. **Strategic alliance / partnership stage**
   At the highest level, things get serious between the buyer and supplier - they're in it for the long haul. This is when they start making big plans together, like investing in each other's businesses, sharing risks, and working on new products or ways of doing things. There's a lot of trust between them, and they've got systems in place that let them work together seamlessly, like shared databases. They're on the same page when it comes to growing and innovating, and they're working together to make it happen.

b) Explain TWO risks of moving too quickly to collaborative relationships. (2 marks)

- **Insufficient trust and capability** – If a company moves too fast to collaboration without building trust and aligning internal processes, suppliers may not share accurate information or may resist changes, leading to frustration and wasted time.
- **High cost and misaligned expectations** – Collaborative and strategic relationships require investment in time, systems, and resources. If expectations are not clearly agreed, one party may feel it is giving more than it receives, which can damage the relationship and reduce benefits.

Question 2 (10 marks)

"Supply chain integration leads to visibility and visibility leads to collaboration."

a) Differentiate between internal integration and external integration. (4 marks)

Internal integration means aligning and coordinating different departments and functions within the same organization, for example, sales, marketing, production, logistics, and finance. The goal is to reduce internal silos so that everyone works with the same data and plans, which improves efficiency and reduces delays.

When a company works with its suppliers, customers, and other partners, it's called **external integration**. This means they share information and make sure their processes, like predicting sales, placing orders, and managing inventory, all work together seamlessly. The goal is to make the whole supply chain run more smoothly. But it's not always easy, because it involves working with other companies and trusting them, and also making sure their computer systems can talk to each other.

b) Explain THREE types of supply chain visibility and their importance. (6 marks)

1. **Supplier visibility**
   Being able to keep an eye on what suppliers are up to, like how much stock they have, when they're producing things, and how well they're doing, is really important. This way, buyers can see if there are going to be any problems, like delays or not having enough stuff, and they can make changes to their plans ahead of time. This helps reduce the chance of running out of things and makes the whole supply chain more reliable. It's like having a heads up, so you can get ready and make adjustments before things go wrong.

2. **Production visibility**
   This refers to real-time information about what is happening in the production process, such as machine status, raw-material availability, and quality checks. Good production visibility allows better planning and faster response to problems, which improves delivery performance and reduces waste.

3. **Inventory visibility**
   Having real-time visibility into inventory levels is a game-changer. It means you can track where everything is, from raw materials to finished products, across all your warehouses and distribution centers. This helps you avoid having too much stock in one place and not enough in another, which can save you money on storage costs. Plus, it makes it easier to get products to customers quickly and accurately, which is a big win. By knowing exactly what you have and where it is, you can make smarter decisions about production, shipping, and more, and that can have a big impact on your bottom line.

Question 3 (10 marks)

A firm is experiencing the bullwhip effect, high inventory cost, and poor service level.

a) Explain THREE causes of the bullwhip effect. (3 marks)

1. **Order batching** – When businesses place large orders infrequently instead of smaller, frequent orders, demand signals become distorted. Upstream suppliers see fewer but larger orders, which makes them think demand is jumping up and down instead of being steady.

2. When companies offer frequent discounts or special deals, it can lead to some pretty big changes in how much customers and distributors buy. They tend to stock up when prices are low, and then slow down their purchases when prices go back up. This creates a bit of a rollercoaster effect, where demand suddenly spikes and then drops, without really being connected to what people actually need or want. As a result, these fluctuations get amplified as you move further up the supply chain, causing even more instability.

3. Inaccurate demand forecasting and information distortion – If each stage in the chain uses its own forecast and does not share real-time sales data with others, small errors are multiplied as they move up the chain. This leads upstream suppliers to overproduce or underproduce compared to actual end-customer demand.

b) Discuss FOUR inventory management strategies that can reduce supply chain cost. (4 marks)

1. **Use of accurate demand forecasting** – By using better forecasting methods and sharing real-time sales data across the chain, firms can hold inventory closer to actual demand, reducing both overstocking and stockouts.
2. I**mplementing Economic Order Quantity (EOQ) models** – EOQ helps find the optimal order size that balances ordering costs and holding costs, so the company avoids placing too many small orders or very large ones that increase either cost.
3. **Adopting just-in-time (JIT) or continuous replenishment** – JIT reduces inventory levels by receiving goods only when they are needed for production or sale. This lowers holding costs and reduces the risk of obsolete stock.
4. **Vendor-managed inventory (VMI)** – In VMI, the supplier monitors the buyer's inventory and replenishes it automatically. This improves coordination, reduces safety stock, and smooths demand signals in the chain.

c) Explain THREE best practices in supply chain management that improve performance. (3 marks)

1. When companies share important information with their suppliers and customers, it really helps everyone work together better. By sharing things like what they think people will buy, how much stock they have, and their plans for making things, everyone can plan more accurately. This also reduces uncertainty and helps improve the service they provide. It's all about working together and sharing information to make things run more smoothly.

2. Continuous improvement and performance measurement – Regularly measuring KPIs (such as on-time delivery, inventory turns, and lead time) and taking corrective actions helps identify problems early and gradually improve efficiency and reliability.

3. Customer-centred planning – Aligning supply chain activities with customer needs (such as product availability, delivery speed, and reliability) ensures that the chain supports the business strategy and builds customer loyalty over time.